Downloaded from gsabulletin.gsapubs.org on October 27, 2010

Geological Society of America Bulletin

Age and origin of Truk Atoll, eastern Caroline Islands: Geochemical, radiometric-age, and paleomagnetic evidence

BARBARA H. KEATING, D. P. MATTEY, J. NAUGHTON and C. E. HELSLEY

Geological Society of America Bulletin 1984;95;350-356
doi: 10.1130/0016-7606(1984)95<350:AAOOTA>2.0.CO;2

Email alerting services click www.gsapubs.org/cgi/alerts to receive free e-mail alerts when new articles cite this article

Subscribe click www.gsapubs.org/subscriptions/ to subscribe to Geological Society of America Bulletin

Permission request click http://www.geosociety.org/pubs/copyrt.htm#gsa to contact GSA

Copyright not claimed on content prepared wholly by U.S. government employees within scope of their employment. Individual scientists are hereby granted permission, without fees or further requests to GSA, to use a single figure, a single table, and/or a brief paragraph of text in subsequent works and to make unlimited copies of items in GSA's journals for noncommercial use in classrooms to further education and science. This file may not be posted to any Web site, but authors may post the abstracts only of their articles on their own or their organization's Web site providing the posting includes a reference to the article's full citation. GSA provides this and other forums for the presentation of diverse opinions and positions by scientists worldwide, regardless of their race, citizenship, gender, religion, or political viewpoint. Opinions presented in this publication do not reflect official positions of the Society.

Notes

Geological Society of America

THE GEOLOGICAL SOCIETY OF AMERICA

Downloaded from gsabulletin.gsapubs.org on October 27, 2010

Age and origin of Truk Atoll, eastern Caroline Islands: Geochemical, radiometric-age, and paleomagnetic evidence

BARBARA H. KEATING
D. P. MATTEY
J. NAUGHTON
C. E. HELSLEY

Hawaii Institute of Geophysics, University of Hawaii, 2525 Correa Road, Honolulu, Hawaii 96822

ABSTRACT

Geochemical, K-Ar age, and paleomagnetic data are presented for volcanic rocks from Truk Lagoon (7°20′N, 151°15′E). Petrographic observations and 52 analyses for major and trace elements reveal 2 shield-building magma types—a mildly alkalic Truk Main Lava Series (TMLS) and a less alkalic Truk Transitional Lava Series (TTLS); and a post-erosional, highly silica-undersaturated, and incompatible-trace-element–enriched Truk Nephelinite Series (TNS). The TMLS and TTLS shield-building lavas give a mean K-Ar age of 10.9 m.y.; post-erosional TNS lavas are dated at 4.8 m.y. B.P. Paleomagnetic results for 15 stably magnetized sites give a paleopole of declination and inclination of 359.3°E and 8.8°30′, respectively, which suggests that Truk was formed 4.4° north of the equator. These data are consistent with a hot-spot origin for the volcanics of Truk Atoll.

INTRODUCTION

Truk Atoll is the westernmost of the only three high volcanic islands in the eastern Caroline Islands, a linear chain of islands, atolls, and seamounts in the western Pacific (see Fig. 1). These seamounts and islands extend eastward from the Mariana trench between the Caroline and Naru basins. Truk Lagoon (7°20′N, 151°15′E) contains numerous small islands representing the highly eroded and submerged remnants of a large shield volcano. The coral reef and associated islets enclosing Truk Lagoon (Fig. 2) presumably trace the former extent of the volcano, which may have had a height of 1,200 m above present sea level.

The general geology of Truk has been described by Stark (1958) and Stark and Hay (1963). Fuller and others (1980) reported paleomagnetic data for samples from 13 localities. We report geochemical, radiometric, and paleomagnetic data for volcanic rocks collected from 22 localities in order to further clarify the tectonic history of the Caroline Islands trend. Bracey and Andrews (1974) suggested that the western Caroline Islands (west of 164 °E) were a relict island arc; Clague and Jarrard (1973) proposed that the Caroline Islands were formed by a melting anomaly or hot spot. The paleomagnetic, radiometric, and geochemical data presented in this paper all support a hot-spot origin.

SAMPLING AND ANALYTICAL METHODS

Rock samples were collected from nine of the islands within Truk Lagoon (Fig. 2). Fifty-two volcanic rocks, consisting of the lithological units sampled for paleomagnetic studies (see below) augmented with samples from additional localities (see Table 1 and Fig. 2), have been studied petrographically and analyzed. Major elements and the trace elements Ni, Cr, Zr, Ga, Rb, Sr, Y, Nb, Ba, La, Ce, Nd, Pb, and Th were determined by X-ray fluorescence spectrometry at the Department of Geological Sciences, University of Birmingham, England. Analytical methods and the trace-element data are reported in Mattey (1982).

Twelve units were selected for potassium-argon age dating. The extraction and analysis systems and the techniques used for K-Ar dating were similar to those described by Dalrymple and Lanphere (1969). Samples were crushed and sieved to 50–100 mesh size (a size that we have found to minimize argon contamination; Keeling and Naughton, 1974), and an aliquot of each sample was ground further for potassium analysis. Potassium was determined by atomic absorption after the samples were dissolved in hydrofluoric acid in a fluorocarbon plastic-lined bomb (Barnas, 1968).

For paleomagnetic study, 4 to 6 oriented samples of each lithological unit were collected from 23 sites. The samples were collected with a portable gasoline-powered drill and oriented by using a sun compass or by back sighting. Using a Schonstedt spinner magnetometer, the natural remanent magnetization (NRM) was then measured for all specimens. Four to eight specimens of each site were chosen for pilot alternating-field (AF) demagnetization behavior studies. All of the specimens were then demagnetized (by using a 3-axis tumbler system within a mu-metal room) between 75 and 400 Oe, depending on the coercivity of the pilot samples. When data from more than one specimen were available, the result from the most stably magnetized specimen from each sample was used for statistical analysis.

RESULTS

Petrography and Geochemistry

A wide variety of rock types from the Truk sample set have been identified by conventional petrographic and chemical criteria. The most common rock types are moderately olivine and plagioclase phyric basalts, ankaramites, and oceanites. The remaining samples consist of hawaiite, nephelinite, mugearite, and a trachyte. Analytical results for representative volcanic rocks from Truk are presented in Table 1. Three magma types were distinguished geochemically by Mattey (1981, 1982). These are named (1) Truk Main Lava Series (TMLS), (2) Truk Transitional Lava Series (TTLS), and (3) Truk Nephelinite Series (TNS). The petrology of these magma types is discussed in Mattey (1982 and unpub. data).

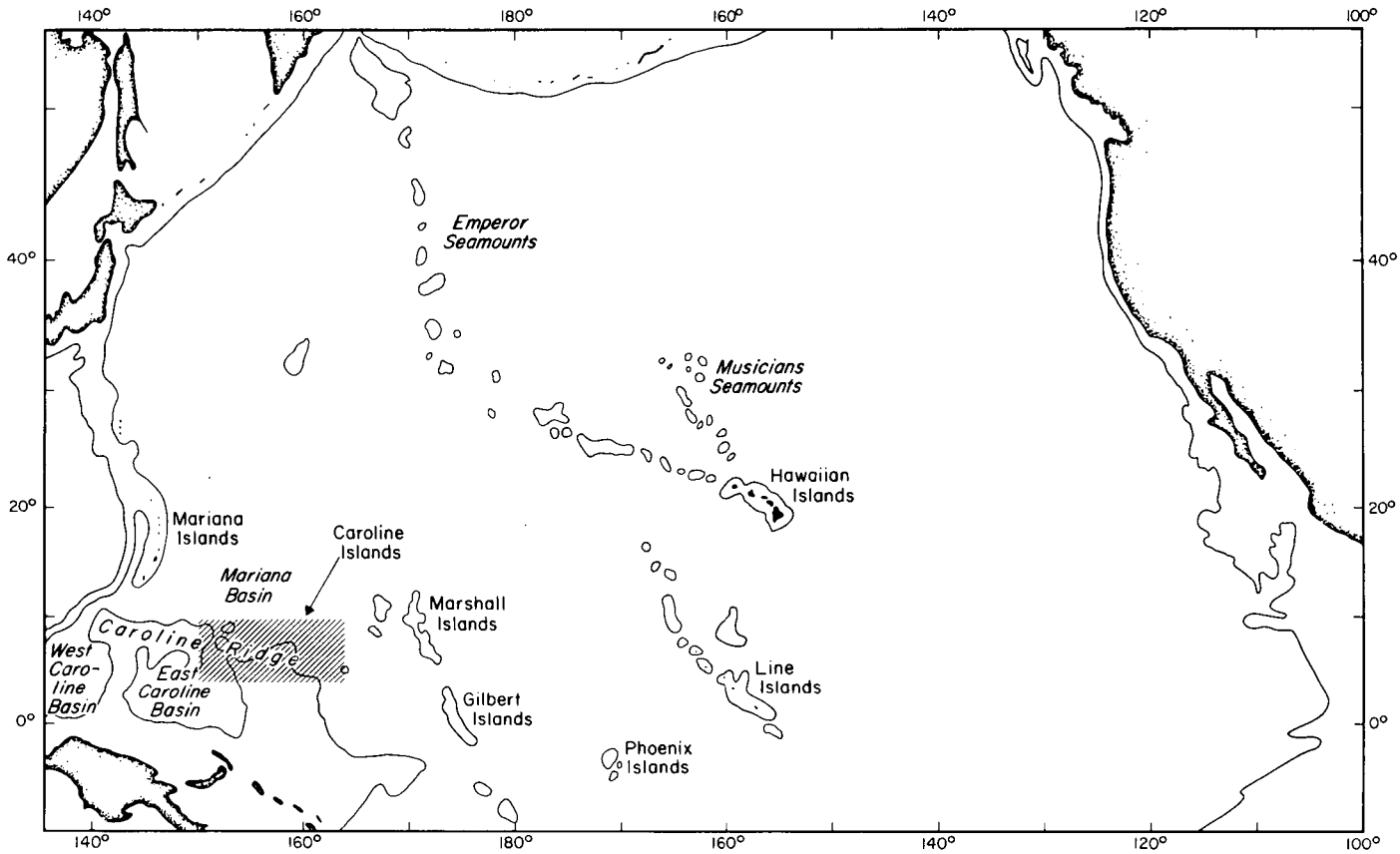
Figure 1. (top) Generalized geographic map of the Pacific Ocean with the position of the Caroline Islands marked by hatched lines. (bottom) Hatched area is shown in detail, including the location of the high volcanic islands of Truk, Ponape, and Kusaie.

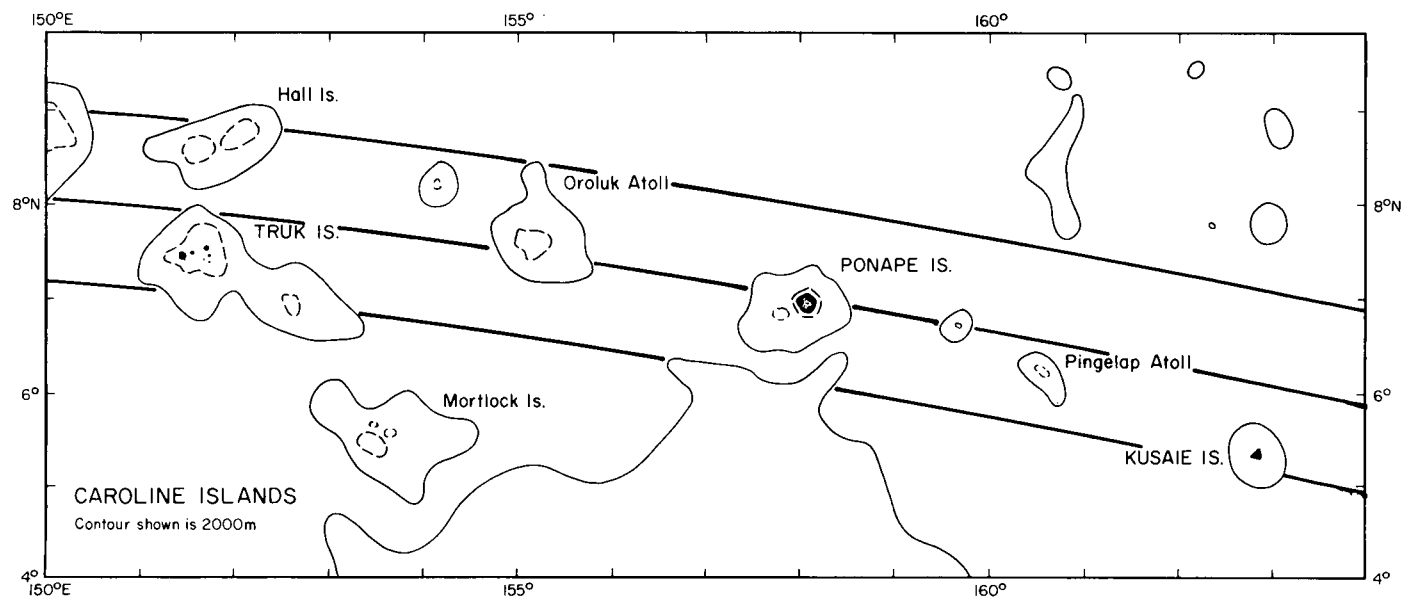
Geological Society of America Bulletin, v. 95, p. 350–356, 6 figs., 4 tables, March 1984.

Downloaded from gsabulletin.gsapubs.org on October 27, 2010

The Truk Main Lava Series

The majority of analyses of rocks collected from Truk Atoll define a comagmatic, mildly alkalic rock series; these are grouped as the Truk Main Lava Series. This magma type is characterized by good linear positive correlations between trace-element pairs such as Nb/Zr (see Fig. 4 below) which fall within the field for Hawaiian alkali basalts on the silica-alkalies diagram (Fig. 3). TMLS basalts are mildly *ne*- to *hy*-normative when Fe_2O_3 is fixed at 15% of total iron and they contain phenocrysts of olivine and plagioclase. The trace- and minor-element variation within the TMLS, discussed in detail by Mattey (1982), suggests that, on a gross scale, element variation is compatible with open-system fractionation of observed pheno-





Downloaded from gsabulletin.gsapubs.org on October 27, 2010

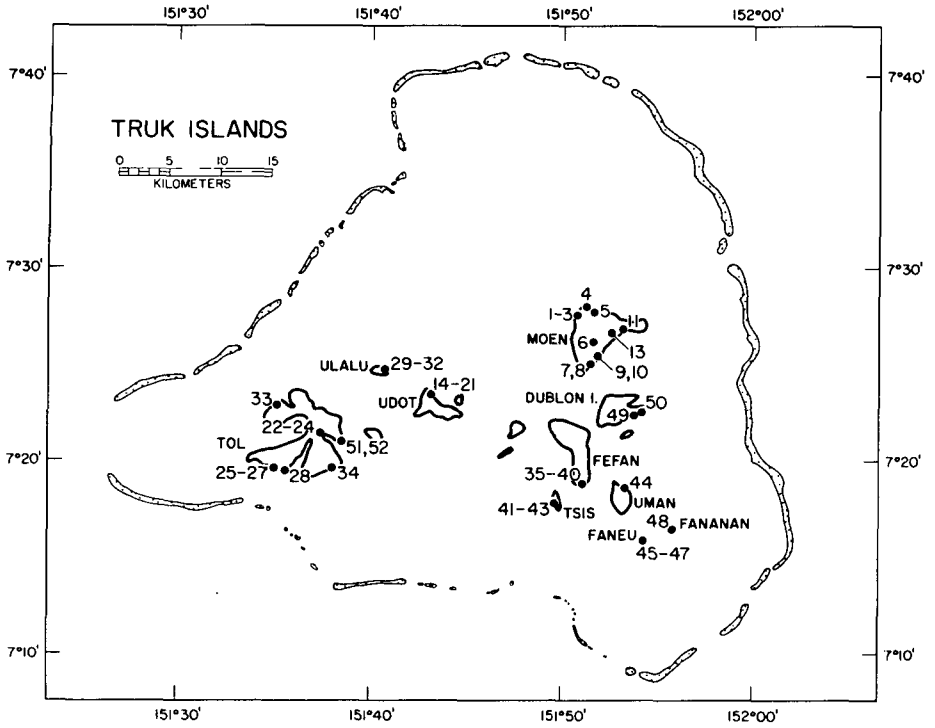


Figure 2. Map of Truk Atoll showing sample localities.

cryst phases from a relatively uniform parental magma.

The Truk Transitional Lava Series

Most of the samples analyzed from Tol Island, located in the western portion of Truk Lagoon (see Fig. 2), can be distinguished geochemically from the TMLS and have been grouped as the Truk Transitional Lava Series. These rocks can be identified principally by slightly lower ratios of elements such as Nb/Zr (Fig. 4) and lower over-all abundance of incompatible minor and trace elements at equivalent MgO or Ni contents (Table 1). TTLS magmas plot across the boundary dividing alkalic from tholeiitic basalts on the silica alkalies diagram (Fig. 3). These dominantly *hy*-normative, olivine and plagioclase phyric TTLS basalts are less "alkalic" than the "Main Lava Series." Mattey (1982) suggested that the TMLS and TTLS may be related by slightly different degrees of melting of a similar mantle composition.

TABLE 1. REPRESENTATIVE ANALYTICAL RESULTS FOR TRUK LAVAS

Magma type	TMLS						TTLS			TNS	
Sample	T39	T42	T49	T40	T12	T3	T13	T28	T23	T31	T52
Site	19	20	24	19	8	1	9	13	11	14	T52
Rock type	Oc	Ank	Bas	Haw	Haw	Mug	Tra	Bas	Haw	Neph	Neph
SiO_2	45.68	46.84	45.58	47.32	51.19	56.74	62.66	48.09	50.78	41.76	42.36
TiO_2	2.64	2.94	3.27	3.95	3.14	2.01	1.04	1.74	3.45	3.54	3.22
Al_2O_3	12.05	14.74	14.15	15.50	16.26	16.85	19.46	17.14	12.92	11.98	11.57
ΣFeO	14.43	12.28	13.23	12.03	10.28	7.05	3.14	10.87	13.21	10.62	9.88
MnO	0.17	0.15	0.15	0.18	0.18	0.20	0.12	0.15	0.15	0.17	0.18
MgO	12.44	9.00	9.14	6.18	4.78	3.52	1.69	7.64	6.69	11.93	11.55
CaO	9.38	9.67	9.70	9.00	7.72	4.95	0.87	10.16	8.22	12.96	14.99
Na_2O	2.29	3.05	3.70	3.67	4.31	6.01	7.21	3.54	3.12	3.63	3.68
K_2O	0.62	0.93	0.60	1.25	1.49	2.07	3.69	0.45	0.97	1.98	1.46
P_2O_5	0.29	0.39	0.49	0.63	0.66	0.60	0.13	0.20	0.48	1.42	1.11
CIPW norms (standard-state Fe_2O_3)											
Q	0.0	0.0	0.0	0.0	0.0	0.0	2.5	0.0	0.9	0.0	0.0
C	0.0	0.0	0.0	0.0	0.0	0.0	2.4	0.0	0.0	0.0	0.0
Or	3.6	5.3	3.4	6.9	8.3	11.9	23.1	2.6	5.7	0.0	0.0
Ab	18.7	21.5	19.5	29.3	34.5	48.9	64.6	24.6	26.1	0.0	0.0
An	20.5	23.0	19.3	20.7	19.5	12.7	4.0	28.3	18.1	10.2	10.9
Lc	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	8.94	6.9
Ne	0.2	1.9	5.5	0.0	0.0	0.0	0.0	2.2	0.0	16.2	17.3
Di	19.1	17.2	19.2	14.4	10.7	6.3	0.0	15.3	15.8	34.0	39.7
Hy	0.0	0.0	0.0	5.3	6.5	8.1	4.4	0.0	21.6	0.0	0.0
Oe	28.3	19.3	18.2	8.9	4.9	0.1	0.0	16.8	0.0	14.8	13.0
Mgt	2.2	2.1	2.8	2.8	2.8	3.6	0.0	2.1	2.9	2.8	3.0
Ilm	4.9	5.4	5.9	7.1	5.6	3.7	1.1	3.2	6.5	6.5	6.3
Ap	0.7	0.9	1.1	1.4	1.2	1.4	0.3	0.5	1.1	3.3	2.7
Trace elements											
Ni	157	211	231	0	0	0	0	69	41	212	237
Cr	437	226	254	0	0	0	0	125	24	311	378
Rb	10	17	6	26	31	29	45	5	17	50	45
Sr	317	432	448	540	560	543	32	334	339	1160	1059
Y	22	25	27	39	46	57	79	24	32	36	31
Zr	153	170	210	255	354	445	655	110	320	365	272
Nb	18	20	26	34	43	65	89	10	30	121	104
Ba	191	231	286	376	543	985	1673	112	299	1002	927
La	16	17	24	30	37	52	63	11	24	61	49
Ce	34	35	49	58	78	113	110	28	53	115	81
Nd	21	23	28	35	56	61	56	17	31	54	39
Pb	4	5	7	5	2	2	8	1	1	8	4
Th	2	4	1	4	4	4	7	1	4	9	10

Notes: Oxide in weight percent normalized to 100% reduced and water free; trace elements in ppm. Normative mineralogy calculated using fixed values of Fe_2O_3 according to Thompson and others (1972). Abbreviations: Oc, oceanite; Ank, ankaramite; Bas, basalt; Neph, nephelinite; Haw, hawaiite; Mug, mugearite; Tra, trachyte.

Downloaded from gsabulletin.gsapubs.org on October 27, 2010

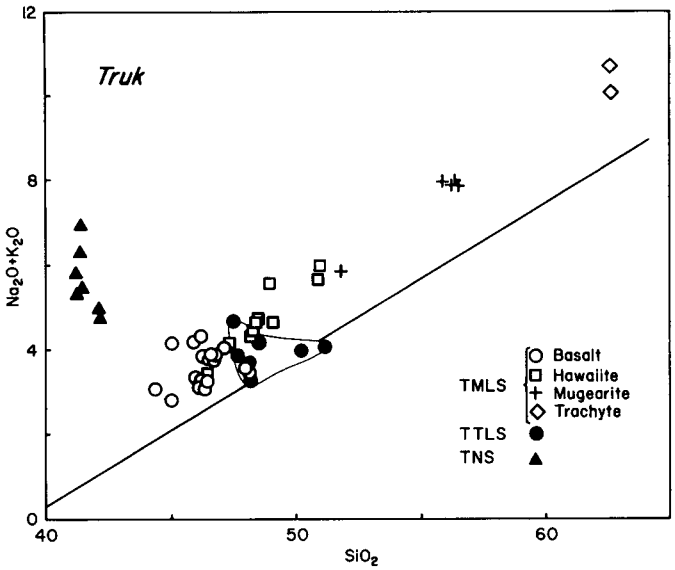


Figure 3. Plot of SiO_2 versus total alkalies for volcanic rocks from Truk. Line divides Hawaiian alkali basalts from Hawaiian tholeiitic basalts (from Macdonald and Katsura, 1964). TTLS, Truk Transitional Lava Series. TMLS, Truk Main Lava Series. TNS, Truk Nephelinite Series.

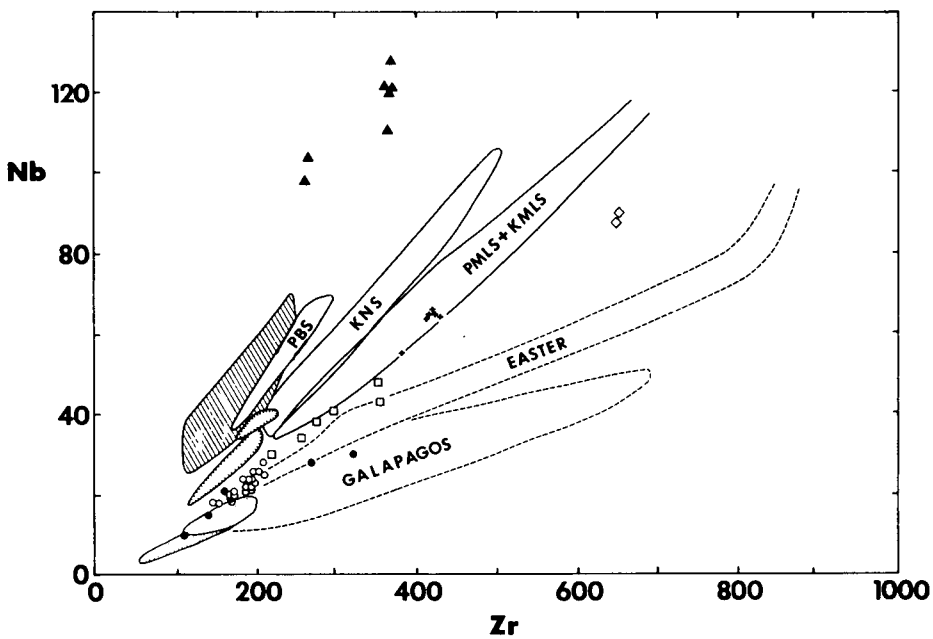


Figure 4. Plot of Nb versus Zr for Truk volcanic rocks (same symbols as Fig. 3). Also plotted for comparison are fields of variation for (1) Hawaiian tholeiites (stippled) (data for Hilina Pali Formation Kilauea from Easton and Garcia, 1980, and for Suiko Seamount from Cambon and others, 1980), alkali basalts (partly hatched) (data for Hualalai volcano from Clague and others, 1980), and post-erosional basalts (hatched) (data for Honolulu Volcanic Series from Clague and Frey, 1982); (2) other Caroline magma types: PMLS, Ponape Main Lava Series; KMLS, Kusaie Main Lava Series; PBS, Ponape Basanite Series (data from Mattey, 1982); and (3) other Pacific ocean island series: Galapagos data from Baitis and Lindstrom (1980); Easter Island data from Baker and others (1974).

The Truk Nephelinite Series

Stark (1958) distinguished nepheline and melilite-nepheline basalts that infill valleys cut into older basalts and dikes on the islands of Tol and Ulalu (Fig. 2). Samples of these post-erosional lavas analyzed in this study contain abundant olivine crystals set in a groundmass of augite, nepheline, magnetite, and rare perovskite. These highly undersaturated (12% to 22% normative nepheline) and magnesian lavas are highly enriched in Rb, Th, Ba, Sr, Nb, La, Ce, Nd, and Zr (Table 1). Significantly higher ratios of Ba/La, Th/La, and Nb/La suggest derivation from a different mantle source relatively enriched in highly incompatible elements (Mattey, 1981, 1982).

POTASSIUM-ARGON AGES

The results of the analyses involved in the dating measurements and the calculated ages are listed in Table 2. The ages were calculated by using the decay constants for ^{40}K recommended by the IUGS Subcommission on Geochronology (Steiger and Jäger, 1977).

We found that in the analysis of potassium, the deviation in the range for duplicate determinations averaged about 3% or less of the value. For some inhomogeneous samples, when the deviation was excessive, three or four replicate determinations were made with average standard deviation of as much as 5%. Because of the difficult and slow argon isotopic measurements, replicates were run in only a few cases. For these, although the reproducibility was poor, mean ages were calculated (Table 2). These are weighted means, where the mean is calculated by weighting the ages according to the inverse of their variances. This practice, followed in many publications on geochronology, is used here for comparative purposes (it has the advantage of allowing the quality of the individual measurements to be taken into account in calculating the mean). The indicated error for each mean (Table 2) is weighted and calculated by the same method.

The low precision in the measurement of the potassium and argon in volcanic rocks has been observed by others (Funkhouser and others, 1968; Doell and Dalrymple, 1973; Dalrymple and others, 1974), and the same poor reproducibility is evident in our results. This scatter for argon is generally ascribed to excess argon present in inclusions, but we are inclined to agree with Doell and Dalrymple (1973) and to assign the low precision for both elements to small-scale inhomogeneities within the samples. Although thin sections of several samples showed evidence of this inhomogeneity as fine-grained segregations and crystal clots, incipient second-

Downloaded from gsabulletin.gsapubs.org on October 27, 2010

TABLE 2. POTASSIUM-ARGON AGES FOR TRUK VOLCANIC ROCKS

Sample	Site	Magma type	K (wt %)		Wt (g)	$\frac{^{40}Ar\ (rad)}{(cc/g \times 10^{-7})}$	$\frac{^{40}Ar\ (rad)}{^{40}Ar\ (tot)} \times 100$	Age (m.y.)	
T35	19	TMLS oceanite	0.615	0.013	7.009	3.458	49.8	14.8	0.4
			0.373	0.017	6.030	1.700	39.5	11.7	0.8
					6.002	1.643	36.3	11.3	1.1
T44	21	TMLS basalt	0.826	0.018	5.495	2.782	51.6	8.4	0.4
					5.473	3.305	55.5	10.5	0.9
					3.187	3.142	26.8	9.9	0.7
T45	22	TMLS basalt	0.481	0.016	4.014	1.824	44.9	9.8	0.5
					4.040	1.958	47.2	10.5	0.5
T48	23	TMLS basalt	0.615	0.013	7.987	2.319	52.8	9.7	0.4
					5.466	2.694	44.2	11.3	0.4
T49	24	TMLS basalt	0.616	0.018	5.141	7.209	42.4	12.7	0.8
					5.600	2.478	38.3	10.4	0.5
T50	25	TMLS basalt	0.753	0.021	5.559	2.923	13.8	10.0	0.9
					5.504	3.130	61.7	10.7	0.9
			0.792	0.002	5.166	4.431	69.1	14.4	0.4
T25	2	TTLS hawaiite	0.622	0.013	5.35	2.257	35.8	9.3	0.6
T28	13	TTLS basalt	0.295	0.014	5.563	0.823	16.9	7.2	1.2
—	15	TTLS(?)	0.817	0.019	5.283	2.736	17.7	8.6	0.8
T33	16	TTLS hawaiite	0.422	0.007	5.002	1.734	23.1	10.6	0.8
			1.330	0.007	6.003	5.309	50.2	10.3	0.8
T31	14	TNS nephelinite	1.312	0.004	5.018	2.730	32.5	5.4	0.2
			1.414	0.004	5.415	2.185	37.3	4.0	0.3
					4.859	2.343	24.6	4.3	0.2
T52	—	TNS nephelinite	0.679	0.013	5.461	1.274	17.3	4.8	0.3

ary zeolite-facies alteration may be the primary cause of sample inhomogeneity. In addition, the segregation of potassium and magmatic argon into the glassy regions or grain boundaries (Mankinen and Dalrymple, 1972) with high dislocation contents would be selective depending on atom or ion size and charge and would be an additional factor producing the observed variation.

Ages obtained for samples from Truk vary from 14.8 m.y. to 4.0 m.y. (Table 2), and the majority of the dikes, sills, and lavas belonging to the TMLS and TTLS fall within the range of 14 m.y. to 7 m.y. TMLS basalts from Fefan and Dublon give the oldest dates, from ~13.9 m.y. B.P. to ~12.4 m.y. B.P., respectively. Geochemically similar samples from Uman (and neighboring isles of Faneau and Fananan) (Fig. 2) are slightly younger at ~9.9 m.y. B.P. (see Table 4 below). A weighted mean age for the TMLS is 12.6 m.y. The interpretation of ages obtained for members of the TTLS presents a problem. Dated examples of this less alkalic magma type (see above) are apparently slightly younger than the TMLS: their ages range from 10 m.y. to 7.2 m.y. (Table 2). TTLS rocks, however, are relatively altered, and sample T28, which gives the youngest age for this magma type with the largest standard deviation (7.2 ± 1.2 m.y.), contains some secondary zeolite. A weighted mean calculated for this magma type gives an average age of 9.9 ± 0.5 m.y., which is similar to the age of TMLS sills dated from Uman (see Table 4 below). The difference in ages (in terms of range) between the two magma types thus may not be significant.

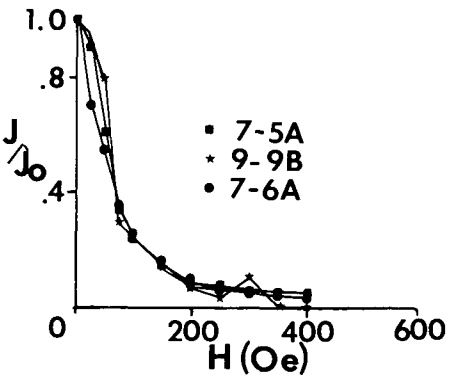


Figure 5. Normalized alternating-field intensity decay curves for representative samples from Truk Lagoon.

The highly silica-undersaturated TNS lavas that uncomformably overlie TMLS and TTLS lavas on Tol and Ulalu and occur as dikes cutting TMLS lavas on Udot give a mean age of 4.8 m.y. TNS magmas are, therefore, at least 3 m.y. younger than the youngest TTLS lavas and 9 m.y. younger than the oldest TMLS rocks dated from Truk.

PALEOMAGNETIC RESULTS

From the 22 sites studied, only 15 rock units appear to have preserved a reliable record of the direction of the paleofield. Most rock units display more scatter than would be expected for single cooling units (flow, dike, or sill) and often have α_{95}'s of more than 10°. The scatter between multiple specimens cut from the same core is also large and suggests that scatter is due to inhomogeneities within individual rock units. Our observations of opaque minerals under reflected light indicate that those rocks having unstable magnetization contain titanomagnetite not only as a groundmass phase but also as large phenocrysts and as partly resorbed xenocrysts often 0.5 to 2 mm in diameter. The latter occur up to 4% by volume, particularly in hawaiites and mugearites, and lack evidence of oxidation or exsolution. Because these grains are not subdivided, their large size produces multidomain grain-size behavior. We suggest that the unstable magnetization results from the large titanomagnetite phenocrysts and xenocrysts that are distributed inhomogeneously within the rock (relative to the size of the paleomagnetic samples). In contrast, sites with rocks that are vesicular or closely jointed are characterized by grains that tend to show evidence of oxidation and varying degrees of exsolution. The rocks of the latter type display relatively high magnetic stability

Downloaded from gsabulletin.gsapubs.org on October 27, 2010

and less scatter in sample directions. Curie temperature analyses of these rocks confirm that titanomagnetite is the carrier of remanence.

The results of the alternating-field demagnetization are shown in Figures 5 and 6. These orthogonal plots illustrate the magnetic behavior of selected samples in peak fields of 400 Oe. With progressive demagnetization, the low-coercivity components of magnetization are removed, and the sample directions trend toward the origin of the plot, indicative of stable magnetization.

In order to determine a reliable paleopole position for Truk, we have attempted to remove sites characterized by unstable magnetization from the Fisherian analysis. The mean site directions after demagnetization are listed in Table 3. (The optimum demagnetization field was defined in these studies by using the Symons and Stupavsky, 1974, stability index.) The mean declination determined for the volcanics of Truk Lagoon is 359.3°E, and the mean inclination is 8.8°N (N = 15, α_{95} = 5.98), indicating that Truk formed about 4.4°N of the equator. Rock units having an α_{95} in excess of 15° have been omitted from the paleopole position calculation. In addition, we have omitted the results from sites 15, 17, and 23, which yielded site directions well removed from the cluster of site means from the other sites.

Although the results from sites 15 and 17 yield relatively well-clustered directions (α_{95} = 7 and 8), we believe that the directions are not valid indications of the axial dipolar field direction. In both cases, the outcrop may not be in place; that is, the outcrop could be a large slump block not recognized in the field because of the thick jungle-vegetation cover. Alternatively, these sites may record directions acquired during excursions or field transitions.

It is interesting to note that site 9, at the summit of Mount Witipon, consists of trachyte, whereas all other sites are mafic in nature. The paleomagnetic directions determined from this exposed site are totally scattered yet apparently stable, a characteristic of rocks that have been remagnetized by lightning strikes.

SUMMARY AND CONCLUSIONS

Paleomagnetic, radiometric, and geochemical studies were conducted on the intrusive and extrusive volcanic units exposed in nine islands in Truk Lagoon, in order to determine the geologic age and origin of the seamount. The results indicate that the seamount was formed between 14 and 8 m.y. B.P., at a position close to the paleo-

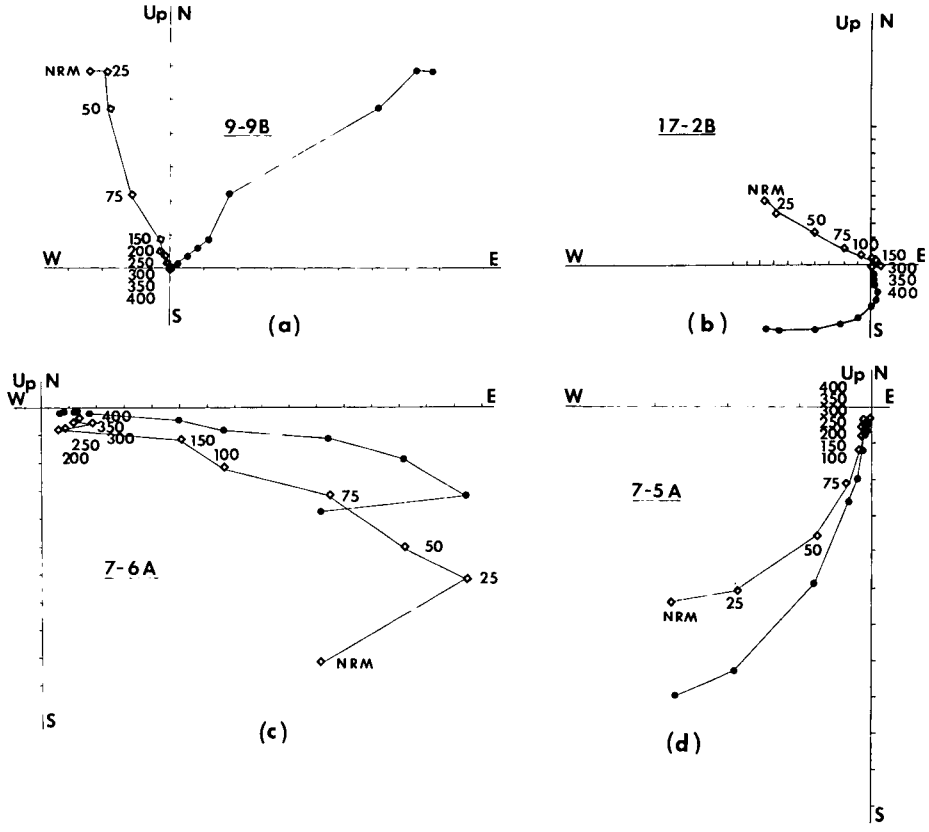


Figure 6. Representative orthogonal AF demagnetization diagrams of successive end points of magnetization vector in peak fields of 400 Oe. The open triangles represent projections on vertical plane. Closed circles represent projections on the horizontal plane. The numbers next to the symbols are the demagnetization steps in Oe.

TABLE 3. SUMMARY OF PALEOMAGNETIC DIRECTIONS FOR SITES IN TRUK LAGOON

Site	Demag. level	Declination	Inclination	Alpha$_{95}$	Polarity	Number of samples
Truk 6	0150	5.9	16.8	2.9	N	6
Truk 7	0150	4.6	16.7	3.7	N	6
Truk 8	0150	2.3	7.7	8.1	N	6
Truk 10	0150	3.5	13.4	8.1	N	40
Truk 11	0200	-18.0	13.2	13.2	R	3
Truk 12A*	0250	2.3	-1.7	11.3	N	4
Truk 12B	0250	-0.3	8.2	7.0	N	7
Truk 13	0200	-0.4	0.2	4.4	N	5
Truk 14	0075	-7.0	5.7	4.5	N	10
Truk 16	0200	-3.4	34.2	4.3	N	5
Truk 19	0150	3.5	11.2	4.3	N	5
Truk 21	0175	-7.4	1.3	16.8	R	3
Truk 22	0050	2.8	5.7	4.5	N	6
Truk 24	0150	-8.9	16.9	7.4	N	7
Truk 25	0200	8.6	-3.0	14.8	N	7
Truk 15†	0250	175.2	42.5	6.8	R	8
Truk 17†	0150	197.7	55.5	8.4	R	6
Truk 26†	0300	4.4	-15.0	23.0	N	4

Declination	Inclination	N§	K**	α_{95}	Dp+	Dm++
359.26°	8.77°	15.0	41.84	5.98	3.64	6.03

Additional sites

Notes: *= Unit A is a dike; unit B is a lava flow. † = Results from these sites were omitted from the pole position determinaton. N§ = Number of sites. K** = Precision parameter. Dp+; Dm++ = Polar error. Samples with reversed polarity have been inverted for Fisherian analysis.

Downloaded from gsabulletin.gsapubs.org on October 27, 2010

TABLE 4. SUMMARY OF AGE DATA FOR TRUK

Magma type	Samples	Island	Mean age
TMLS	T35	Fefan	13.9 ± 0.5
TMLS	T44,T45,T48	Uman	9.9 ± 0.7
	T49,T50	Dublon	12.4 ± 0.7
	Mean		12.6
TTLS	T25,28,33 (+Site 15)	Tol	9.9 ± 0.8
TNS	T31	Ulalu	4.7 ± 0.2
	T52	Tol	4.8 ± 0.2
Shield-building lavas (TMLS + TTLS)			10.9 ± 1.2
Post-erosional lavas (TNS)			4.8 ± 0.2

equator. Our paleolatitude (4.4°N) is in excellent agreement with the 3° of northward motion predicted for the site by using the motion of the Pacific plate about the Hawaiian hot-spot rotation pole at a rate of 1.3°, as suggested by Clague and Jarrard (1973). These results, however, are inconsistent with an island-arc mode of origin for the western Caroline Islands (Bracey and Andrews, 1974) because recent paleomagnetic studies of island-arc volcanics in the western Pacific by Keating and others (1983) and Falvey (1978) indicate that western Pacific island arcs are characterized by very large tectonic rotations.

Weighted mean ages of rocks from Truk magma types are summarized in Table 4. Two major phases of volcanic activity are seen from geologic evidence and from the radiometric age data. A "shield-building" phase distinguished by the mildly alkalic and transitional magmas of the TMLS and TTLS lasted from 14 m.y. B.P. to 8 m.y. B.P., with a weighted mean age of 10.9 m.y. After an erosional interval, activity resumed and highly undersaturated melilite-nephelinite and nephelinites of the TNS were erupted about 4.8 m.y. ago. The volcanic evolution of Truk, therefore, is strongly reminiscent of the evolution of Hawaiian volcanoes characterized by a shield-building phase followed by minor post-erosional activity.

Truk shield-building lavas have been divided into two magma types that may reflect variable degrees of melting of the same mantle source (Mattey, 1982). The Truk Transitional Lava Series, the "least alkaline" of the Truk magma types, straddles the boundary dividing Hawaiian alkalic and tholeiitic lavas (Fig. 3) and is characterized by a mean Nb/Zr ratio (Fig. 4), which is intermediate between those of Hawaiian tholeiites and Hawaiian alkali basalts.

The TMLS, the dominant shield-building magma type on Truk, consists of a differentiated alkalic series more similar to the Hawaiian alkalic series (Figs. 3 and 4). TTLS magmas are restricted to the western portion of Truk Lagoon (Tol Island; see Fig. 2), whereas TMLS magmas have a wider distribution and are found on both Tol and the eastern islands of Moen, Dublon, Fefan, and Udot (Fig. 2). The K-Ar age data suggest that TTLS magma production was coeval with the later stages of TMLS eruption. After an erosional interval of at least 3 m.y., highly silica-undersaturated and trace-element–enriched nephelinites were erupted over eroded TTLS and TMLS lavas on Tol and Ulalu. These "post-erosional" TNS lavas were derived from a source more enriched in Ba, Nb, K, Rb, La, Ce, and Nd than the source for TMLS and TTLS magmas (Mattey, 1982). Compared to the Honolulu Volcanic Series of Oahu (Clague and Frey, 1982), the TNS mgmas at equivalent MgO and Ni contents and similar silica undersaturation are more enriched in Nb and Zr than are their Hawaiian counterparts (Fig. 4).

The petrography and geochemical evolution of Truk, therefore, is strongly reminiscent of that of islands in the Hawaiian chain; however, the "shield-building" lavas of Truk are compositionally more similar to the alkalic lavas that typically form only thin, late-stage caps on many Hawaiian volcanoes. Nevertheless, the combined geophysical and geologic data from the Caroline Islands are consistent with a "hot-spot" origin for Truk Atoll. The paleomagnetic data—this study and Fuller and others (1980)—give a paleolatitude consistent with accepted plate-motion models. The geochronological studies show an increase in age from Kusaie (~1–3 m.y.) to Ponape (5–7 m.y.) to Truk (~11 m.y.), and the Caroline Islands chain lies along a small circle around the plate–hot-spot rotation pole fitting the Hawaiian chain for 1 to 14 m.y. (Clague and Jarrard, 1973; Jarrard and Clague, 1977; Epp, 1978; Turner and others, 1980). The available data thus are consistent with an origin and geologic evolution similar to that of the Hawaiian Islands chain.

ACKNOWLEDGMENTS

This work was supported by National Science Foundation grant EAR 78-23841 and by the National Environment Research Council (United Kingdom). We thank the Honorable Erhart Aten, Governor of Truk, and his staff for assisting this project and providing logistical support. In addition, we thank Phillip Utjto for his assistance in collecting samples and for leading us to various outcrops. Josh Thompson measured all of the samples for the paleomagnetic study and Andy Balogh completed the computer analysis. Chemical analyses were kindly provided by Dr. G. L. Hendry, University of Birmingham, United Kingdom.

REFERENCES CITED

Baitis, H. W., and Lindstrom, M. M., 1980, Geology and Petrology of Pinzon Island, Galapagos Islands: Contributions to Mineralogy and Petrology, v. 79, p. 367–386.

Baker, P. E., Buckly, F., and Holland, J. G., 1974, Petrology and geochemistry of Easter Island: Contributions to Mineralogy and Petrology, v. 44, p. 85–100.

Barnas, B., 1968, A new method for decomposition and comprehensive analysis of silicates by atomic absorption spectrometry: Analytical Chemistry, v. 40, p. 1682–1686.

Bracey, D. R., and Andrews, J. E., 1974, Western Caroline Ridge: Relic island arc?: Marine Geophysical Researches, v. 2, p. 111–125.

Cambon, P., Joron, J. L., Bougault, H., and Trevil, M., 1980, Leg 55, Emperor Seamounts: Trace elements in transitional tholeiites, alkali basalts and hawaiites—Mantle homogeneity or heterogeneity and magmatic processes, *in* Jackson, E. D., and others, Initial reports of the Deep Sea Drilling Project, Volume 55: Washington, D.C., U.S. Government Printing Office, p. 585–597.

Clague, D. A., and Frey, F. A., 1982, Petrology and trace element geochemistry of the Honolulu Volcanic Series, Oahu: Implications for the mantle below Hawaii: Journal of Petrology, v. 23, p. 447–504.

Clague, D. A., and Jarrard, R. D., 1973, Tertiary plate motion deduced from the Hawaiian-Emperor Chain: Geological Society of America Bulletin, v. 84, p. 1135–1154.

Clague, D. A., Jackson, E. D., and Wright, T. L., 1980, Petrology of Hualalai volcano, Hawaii: Implications for mantle composition: Bulletin Volcanologique, v. 43–44, p. 641–656.

Dalrymple, G. B., and Lanphere, M. A., 1969, Potassium-argon dating: Principles, technique and application to geochronology: San Francisco, Freeman and Company, p. 1–258.

Dalrymple, G. B., Lanphere, M. A., and Jackson, J. D., 1974 Contribution to the petrography and geochronology of volcanic rocks from the Leeward Hawaiian Islands: Geological Society of America Bulletin, v. 84, p. 727–738.

Doell, R. R., and Dalrymple, G. B., 1973, Potassium-argon ages and paleomagnetism of the Waianae and Koolau Volcanic Series, Oahu, Hawaii: Geological Society of America Bulletin, v. 84, p. 1217–1242.

Easton, R. M., and Garcia, M. O., 1980, Petrology of the Hilina Formation, Kilauea volcano, Hawaii: Bulletin Volcanologique, v. 43–44, p. 657–673.

Epp, D., 1978, Age and tectonic relationships among volcanic chains on the Pacific plate [Ph.D. dissert.]: Honolulu, Hawaii, University of Hawaii, p. 1–199.

Falvey, D. A., 1978, Analysis of paleomagnetic data from the New Hebrides: Bull. Aust. Soc. Explor. Geophys., v. 9, no. 3, p. 117–123.

Fuller, M., Dunn, J. R., Green, G., Lin, Jin-Lu, McCabe, R., Torey, K., and Williams, I., 1980, Paleomagnetism of Truk Islands, eastern Carolines and of Saipan, Marianas: American Geophysical Union Geophysical Monograph 22, p. 235–245.

Funkhouser, J. G., Barnes, I. L., and Naughton, J. J., 1968, The determination of a series of ages of Hawaiian volcanoes by the potassium-argon method: Pacific Science, v. 22, p. 369–372.

Jarrard, R. D., and Clague, D. A., 1977, Implications of Pacific island and seamount ages for the origin of volcanic chains: Review of Geophysics and Space Physics, v. 15, p. 57–76.

Keating, B. H., Kodama, K., and Helsley, C. E., 1983, Paleomagnetic studies of the Bonin and Mariana Island arcs, *in* Arc volcanism, physics and tectonics, Advances in Earth and Planetary Science Series, p. 245–260.

Keeling, D. L., and Naughton, J. J., 1974, K-Ar dating: Addition of atmospheric argon on rock surfaces from crushing: Geophysical Research Letters, v. 1, p. 43–46.

Macdonald, G. A., and Katsura, T., 1964, Chemical composition of Hawaiian lavas: Journal of Petrology, v. 6, p. 82–133.

Mankinen, E. A., and Dalrymple, G. B., 1972, Electron microprobe evaluation of terrestrial basalts for whole-rock K-Ar dating: Earth and Planetary Science Letters, v. 17, p. 89–94.

Mattey, D. P., 1981, Secular geochemical variation in lavas from Truk, Ponape and Kusaie, eastern Caroline Islands [abs.]: EOS (American Geophysical Union Transactions) v. 62, p. 1068.

——— 1982, Minor and trace element geochemistry of volcanic rocks from Truk, Ponape, and Kusaie, eastern Caroline Islands evolution of a young hotspot trace across old oceanic crust: Contributions to Mineralogy and Petrology, v. 80, p. 1–13.

Stark, J. T., 1958, Military geology of Truk Islands, Caroline Islands: U.S. Army Chief of Engineers, Intelligence Division, Headquarters, U.S. Army Pacific, 205 p.

Stark, J. T., and Hay, R. L., 1963, Geology and petrology of the volcanic rocks from Truk Islands, east Caroline Islands: U.S. Geological Survey Professional Paper 409.

Steiger, R. H., and Jäger, E., 1977, Subcommission on Geochronology: Convention on the use of decay constants in geo- and cosmochronology: Earth and Planetary Science Letters, v. 36, p. 359–362.

Symons, D. T., and Stupavsky, M., 1974, A rational paleomagnetic stability index: Journal of Geophysical Research, v. 79, p. 1718–1720.

Thompson, R. N., Esson, J., and Dunham, A. C., 1972, Major element chemical variation in Eocene lavas of the Isle of Skye, Scotland: Journal of Petrology, v. 13, part 2, p. 219–253.

Turner, D. L., Jarrard, R. D., and Forbes, R. B., 1980, Geochronology of the Pratt-Welker seamount chain, Gulf of Alaska: A new pole of rotation for the Pacific plate: Journal of Geophysical Research, v. 85, p. 6547–6556.

MANUSCRIPT RECEIVED BY THE SOCIETY APRIL 7, 1982
REVISED MANUSCRIPT RECEIVED MARCH 9, 1983
MANUSCRIPT ACCEPTED MARCH 16, 1983
HAWAII INSTITUTE OF GEOPHYSICS CONTRIBUTION NO. 1437

Printed in U.S.A.